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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: Protein preparation and label-free MS Fresh tissues from maize seedlings (2 weeks old) and the leaves of Arabidopsis (mature), rice (V2 and booting stages), and maize (14 days old) were ground into a powder in liquid nitrogen.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
The aboveground tissues of 2-week-old seedlings (at least six plants per replicate) were collected and ground into a powder for RNA-seq, full-length ribosome profiling (FL_Ribo-seq), 3Ribo-seq, and protein MS.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
A gradient profiler (BioComp, http:www.biocompinstruments.comindex.php) with an EM-1 UV data acquisition system (Bio-Rad, http:www.bio-rad.com) was employed for component analysis according to the manufacturers instructions.	BioComp		
The 3Ribo-seq reads were aligned against these sequences with Bowtie 2 software to remove the rRNA and tRNA reads (Langmead and Salzberg , 2012) .	Bowtie 2		Suggestion: Bowtie 2 (RRID:SCR_016368)(link)
The reads from maize , Arabidopsis , and rice produced by RNA-seq or FL_Ribo-seq were also aligned to the maize reference genome (Zea_mays.AGPv4) , the Arabidopsis reference genome (TAIR10) , and the rice reference genome (Oryza_sativa.IRGSP-1.0) , respectively , using RSEM-1.3.1 (Li and Dewey , 2011) combined with the Bowtie 2 aligner (Langmead and Salzberg , 2012) .	Bowtie		Suggestion: Bowtie (RRID:SCR_005476)(link)
Translatomic annotation of the genome Based on translatomic data from multiple tissues , the Arabidopsis , rice , and maize genomes were re-annotated using the following steps: first , the single-end reads were mapped to the reference genome using HISAT2 (v2.0.4) software (Kim et al. , 2015) with the parameter dta .	HISAT2		Suggestion: HISAT2 (RRID:SCR_015530)(link)
Second , the alignments were assembled into potential transcripts using StringTie (v1.3.0) software (Pertea et al. , 2015) with the parameter G , which produced known and new transcripts .	StringTie		Suggestion: StringTie (RRID:SCR_016323)(link)
The extracted sequences of the new genes were subjected to BLAST analysis against the UniProt protein database , and GO annotation was performed using TBtools (Chen et al. , 2020) and WEGO (v 2.0 , http:wego	BLAST		Suggestion: WU-BLAST (RRID:SCR_011824)(link)
	UniProt		
Population structure and family relatedness were calculated using EMMAX and ADMIXTURE	ADMIXTURE		Suggestion: ADMIXTURE (RRID:SCR_001263)(link)

(Alexander et al. , 2009) , respectively .			
Haploview (Barrett et al., 2005) was used to estimate the correlation between SNPs to filter out cases with high linkage disequilibrium (LD) between markers.	Haploview		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Significant associations were determined based on Bonferroni-corrected p < 8.03 x 10-8 (1N , N = 12 452 367)	Bonferroni-corrected
Oligonucleotides	
Identification of the polyA signal motif Sequences downstream of the stop codons of genes were extracted, and the polyA signal motif was identified with Dragon PolyA Spotter (DPS, v 1.200, https:www.cbrc.kaust.edu.sadpsCapture.html) (Kalkatawi et al., 2013) to identify 6-bp sequence variants with the sequences AATAAA, AAAAAG, AAGAAA, AATACA, AATAGA, AATATA, ACTAAA, AGTAAA, ATTAAA, CATAAA, GATAAA, or TATAAA.	AATAAA
	AAGAAA
	AATAGA
	AGTAAA
	CATAAA

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq